

Solution Requirements

Date	14 July 2026
TeamID	SWUID2026221635
ProjectName	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system must allow users (policymakers/researchers) to securely register and log in using their email and passwords.	Medium
2	Authorization levels	Define two distinct roles: 'Admin' (can upload new datasets and retrain the model) and 'Standard User' (can run predictions and view reports).	Medium
3	External interfaces	The system should be able to pull latest development indexes occasionally using external public APIs (like World Bank API or UN Open	Low

		Data).	
4	Transactions processing	The backend must process user inputs, convert them into array structures, run prediction calculations via the ML model, and save records securely.	High
5	Reporting	Generate interactive visual dashboards using Matplotlib/Seaborn showing development gap charts and allowing PDF report exports	High
6	Businessrules, etc.	Predictions must restrict input metrics to logical constraints (e.g., Lifespan must be between 0 to 120 years, schooling between 0 to 25 years).	High
7	Compliance to laws or regulations	Compliance with general data protection rules (GDPR) ensuring no personally identifiable user data is public or sold.	Medium
8	Other	Analytical outputs must be displayed side-by-side with global baselines using generated trend charts and allow printable summary downloads.	Low

Step2:Non-FunctionalRequirements(NFR)

S.No	NFRCategory	RequirementDescription	Target Metric / AcceptanceCriteria
1	Performance& Speed	Backend processing of ML predictions and UI rendering must be fast and seamless.	ML Prediction loads in \$< 1.5\$ seconds; Web page loads in \$< 2\$ seconds.
2	Scalability	The web server must handle parallel assessment requests without lagging.	Supports up to 1,000 concurrent active sessions.
3	Security&Data Privacy	Protect user account credentials and secure backend databases.	Use SHA-256 for hashing passwords and SSL/TLS encryption for HTTPS data flow.
4	Reliability& Availability	The application must remain online and functional during peak traffic times.	Minimum of 99.5% uptime per month.
5	Usability &Accessibility	The UI must be highly intuitive, clean, and accessible for diverse users	Fully responsive design matching WCAG 2.1 AA accessibility standards for text readability.
6	Other	The system must be fully compatible with multi-platform environments and cross-browser layouts, ensuring consistent UI presentation for all users.	Low